[METRIC]

MIL-I-51324A(EA)
28 May 1980
SUPERSEDING
MIL-S-51324(MU)
16 September 1969

MILITARY SPECIFICATION

INSECTICIDE, SMDC (SODIUM N-METHYLDITHIOCARBAMATE) (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solution of sodium N-methyldithiocarbamate in water (hereinafter referred to as SMDC).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71 - Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)

FSC 6840

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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PPP-C-1337 - Drum, Shipping, Steel, With Polyethylene Insert
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 Gallons)

MILITARY

MIL-V-12276 - Varnish, Phenolic, Baking

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act
49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "40 CFR 162" or "49 CFR 171 to 179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. SMDC shall be a clear, colored, uniform solution having no evidence of suspended matter or sediment when tested as specified in 4.2.4.1.

3.2 Composition. SMDC shall be a solution of sodium N-methyldithiocarbamate in water. The only other permissible optional ingredient shall be a quantity of sodium hydroxide necessary to provide a pH of 9 to 11. When sodium hydroxide is used, the SMDC shall have a pH of 9 to 11 when tested as specified in 4.2.4.2. The contractor shall certify that the SMDC contains no other ingredients.

3.3 Specific gravity. SMDC shall have a specific gravity of no less than 1.160 and no more than 1.180 at 20°/20°C when tested as specified in 4.2.4.3.

3.4 Assay. SMDC shall have an assay as sodium N-methyldithiocarbamate of no less than 32.7 percent by weight and no more than 34.9 percent by weight when tested as specified in 4.2.4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the SMDC produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table I. A representative specimen of approximately 0.5 liter shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE I. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container ready for shipment. Sample shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container interior coating missing or not as specified (where required)
(d) Container closure not as specified
(e) Container damaged or leaking
(f) Marking incorrect, missing, or illegible
(g) Unitization not as specified

4.2.3.2 For test. Each specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

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4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine approximately 50 milliliters (ml) of the specimen for uniformity, clarity, suspended matter, and sediment.

4.2.4.2 pH (when the SMDC contains sodium hydroxide). Determine the pH of the specimen at room temperature by means of a pH meter accurate to 0.1 pH units.

4.2.4.3 Specific gravity. Determine the specific gravity of the specimen at 20°/20°C by any method accurate to the fourth decimal place.

4.2.4.4 Assay.

(a) Preparation of pure SMDC. Extract commercial SMDC three times with four times the volume of ethyl acetate for each extraction. Discard the ethyl acetate extracts. Pour the remaining SMDC solution into a crystallizing dish and blow a stream of air across it until the water has evaporated and crystalline SMDC remains. Freeze dry the crystals in a refrigerator or in a vacuum desiccator containing concentrated sulfuric acid. Use the pure SMDC immediately.

(b) Preparation of calibration curve. Weigh to the nearest 0.1 milligram (mg) approximately 1.0 gram (g) of pure SMDC prepared as specified in (a) and dissolve in 250 ml of water in a volumetric flask. Transfer a 5-ml aliquot to a 100-ml volumetric flask and dilute to the mark with water. Dilute 1, 2, 3, 4, and 5 ml of this solution to the mark with water in five separate 100-ml volumetric flasks. The concentrations of these solutions will be 2, 4, 6, 8, and 10 micrograms per milliliter, respectively. Determine the absorbance of each solution at 281 nanometers on a UV spectrophotometer using water in the reference cell. Prepare a calibration curve by plotting absorbance versus concentration of each solution on regular graph paper.

(c) Procedure. Transfer 1 ml of specimen to a 1000-ml volumetric flask and dilute to the mark with water. Transfer a 2-ml aliquot of this solution to a 100-ml volumetric flask and dilute to the mark with water. (This solution will contain approximately 6.6 micrograms per milliliter.) Determine the absorbance of the solution at 281 nanometers on a UV spectrophotometer using the same conditions that were used to prepare the calibration curve in (b). Read the concentration of the solution from the calibration curve prepared as specified in (b). Calculate the percent assay as follows:

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$$\text{Percent assay} = \frac{5A}{B}$$

where: A = Concentration of specimen solution read from calibration curve in micrograms per milliliter, and
B = Specific gravity determined in 4.2.4.3.

5. PACKAGING

5.1 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.1.1 Level A. A quantity of 18.9 ± 0.1 liters of SMDC shall be packed in a steel pail conforming to type I, class 4 of PPP-P-704. The interior surface of the pail shall be completely lined with two coats of phenolic baking varnish conforming to type II of MIL-V-12276. The thickness of the lining shall be no less than 0.036 millimeters. The pail shall be furnished with a pouring spout fitted to the nozzle and a tamper-proof closure seal.

5.1.2 Level B. A quantity of 18.9 ± 0.1 liters of SMDC shall be packed in a steel drum with polyethylene insert conforming to type I, class 1 of PPP-C-1337.

5.1.3 Commercial. A quantity of 18.9 ± 0.1 liters of SMDC shall be packed commercially in accordance with MIL-STD-1188.

5.2 Unitization. Level A and B packs shall be unitized in accordance with the applicable palletization requirements of MIL-STD-147 using a softwood pallet conforming to type IV of NN-P-71. The load shall be secured to the pallet by strapping using cap and tray, by shrink wrap, or by stretch wrap. Commercial packs shall be palletized to meet applicable common carrier requirements.

5.3 Marking. Level A and B packs and pallet loads shall be marked in accordance with MIL-STD-129. Commercial packs shall be marked in accordance with MIL-STD-1188. Commercial pallet loads shall be marked to assure carrier acceptance.

5.3.1 Labeling. All containers shall be labeled in compliance with 40 CFR 162 and 49 CFR 171 to 179. Labels shall be approved and registered in accordance with 40 CFR 162. As a minimum, each container shall be labeled with the following information:

(a) The name and percent by weight of each active ingredient
(b) The name and percent by weight of each inactive ingredient

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(c) National stock number, military specification number, contract number, and lot number
(d) Manufacturer's name and address
(e) EPA registration and establishment number
(f) Precautionary information
(g) Disposal data
(h) Directions for use:

FOR MILITARY USE ONLY

DIRECTIONS FOR USE FOR TREATMENT

OF SOIL FROM RETROGRADE CARGO

Contaminated soil and waste water from washable cargo will be collected in a pit dug adjacent to the decontaminating facility. Accumulation of material in the pit will be treated with 9.5 liters (2.5 gallons) of SMDC in 568 liters (150 gallons) of water per 84 square meters (100 square yards) of surface area.

Cargo that cannot be treated as above will be sprayed with the above SMDC solution until all the soil is thoroughly saturated, provided that the soil accumulation on the cargo does not exceed 12.7 millimeters (1/2 inch). Cargo thus treated shall be marked for guidance of subsequent receivers.

Soil contaminated cargo that does not readily lend itself to washing shall be mechanically cleaned. The accumulated dirt shall be collected and deposited in the collection pit or treated until saturated at the rate of 3.8 liters (1 gallon) of SMDC to 227 liters (60 gallons) of water.

SMDC-treated waste shall not be discharged into sewer systems or streams.

SMDC and its solutions are corrosive to brass and bronze.

Normally, discharge of soil and washings directly into sea water without SMDC treatment is an acceptable disposal method for agricultural quarantine purposes. The selected discharge points shall be made with the approval of a representative of the Plant Quarantine Division, US Department of Agriculture.

6. NOTES

6.1 Intended use. SMDC is intended for use in the treatment of soil resulting from the cleaning of retrograde cargo.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-51324A(EA) Insecticide, SMCD (Sodium N-Methyldithiocarbamate)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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